Tube-Tightened Bilinear Koopman Model Predictive Guidance for Impact-Angle-Constrained Interception Without Prescribed Impact Time

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***Abstract*—This paper investigates a Koopman-operator-based model predictive guidance method for planar stationary-target interception with an impact-angle constraint but without a prescribed impact-time constraint. Removing the fixed terminal-time requirement changes the guidance objective from scheduled interception to capture with a desired terminal direction, which is less restrictive and more consistent with reachable-set limitations observed in fixed-time tuning. The development separates the unconstrained guidance core from the angle-constrained terminal task. It is first shown that the ideal Cartesian acceleration-controlled stationary-target model admits an exact finite-dimensional continuous-time bilinear Koopman representation under the constant-speed assumption. A geometric terminal-error model is then constructed in the desired impact-angle frame, where along-track progress, cross-track displacement, and terminal heading error provide the regulated prediction outputs. Based on this structure, a task-oriented bilinear extended dynamic mode decomposition predictor is identified from sampled trajectories and embedded into a sequential convex Koopman model predictive controller with a first-order autopilot state, input-difference regularization, terminal slack variables, and validation-data residual-tube tightening. Numerical simulations show that the proposed angle-only design satisfies a 5 m capture-radius requirement for the nominal 30° case, achieving a 3.048 m miss distance and a 0.109° impact-angle error, while the speed-perturbed stress case also remains inside the capture set with a 4.150 m miss distance and a 0.121° angle error. Additional nominal tests satisfy 10 m capture-radius requirements for 30° and 45° impact-angle commands. Acceleration-rate scans show that moderate command-smoothing constraints can improve the capture result, whereas large-angle cases expose the remaining reachable-envelope and solver-reliability limitations. These results support a bounded claim: bilinear Koopman-MPC is a useful QP-based predictive-guidance framework for nominal impact-angle-constrained interception when impact time is treated as an outcome rather than as a hard command.**

***Index Terms*—Koopman operator, model predictive guidance, impact-angle constraint, residual tube, bilinear predictor, angle-only guidance.**

I. Introduction

TERMINAL guidance often requires a pursuer not only to reduce miss distance, but also to approach the target from a prescribed direction. Impact-angle guidance is therefore important in planar interception, precision engagement, and terminal path-shaping problems. Unlike simultaneous impact-time and impact-angle guidance, the angle-only problem does not require the pursuer to arrive at a specified clock time. The controller may instead use the available maneuvering authority to shape the trajectory and let the actual impact time emerge from the closed-loop motion.

Model predictive control (MPC) is attractive for angle-constrained guidance because it can handle terminal tolerances, acceleration bounds, and command-smoothing requirements in a single optimization problem. Direct nonlinear MPC, however, can be too expensive or unreliable for repeated terminal-guidance updates. Koopman operator methods offer an alternative route: nonlinear dynamics are embedded in a lifted observable space so that prediction becomes approximately linear or structured bilinear. Koopman predictors have been used in MPC for nonlinear systems and in Koopman-operator-based integrated guidance and control for high-speed missiles [3], [4], [6]. This paper follows that predictor-then-MPC logic, but narrows the terminal task to impact-angle-constrained capture without enforcing a scheduled impact time.

The main technical issue is that interception is not a regulation equilibrium in the original Cartesian relative coordinates. At capture, the relative position is near zero, but the velocity does not vanish. A useful predictive-guidance coordinate must therefore describe terminal geometry rather than regulate the original position state as a static equilibrium. For the angle-only problem, we use an impact-frame terminal-error state that contains along-track progress, cross-track displacement, and heading error. The along-track component is used as a progress and terminal-position reference, but not as a hard clock-time command.

The main contributions are summarized as follows.

1) A finite-dimensional continuous-time bilinear Koopman representation is derived for the unconstrained constant-speed stationary-target guidance core in Cartesian relative coordinates. This result explains why a bilinear lifted predictor is structurally better matched to the control-affine guidance dynamics than a purely linear input predictor.
2) The terminal guidance task is reformulated as impact-angle-constrained capture in a desired-impact-frame error coordinate. The formulation removes the prescribed impact-time constraint while retaining a terminal target for cross-track error, heading error, and near-target capture.
3) A data-driven sequential convex bilinear Koopman-MPC controller is constructed with measured-predicted lifted-state interpolation, move and acceleration-difference reg-

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ularization, terminal slack variables, and validation-data residual-tube tightening.

4) Numerical simulations report nominal and stress angle-only capture results, prediction-accuracy diagnostics, acceleration-rate constraint scans, and large-angle boundary cases. The results are presented as evidence for nominal angle-only guidance rather than as evidence for fixed-time interception.

The remainder of this paper is organized as follows. Section II formulates the planar guidance core. Section III derives its exact continuous-time bilinear Koopman structure. Section IV introduces the angle-only terminal-error model. Section V presents the bilinear EDMD predictor and sequential convex MPC design. Section VI describes residual-tube terminal tightening. Section VII reports numerical results, and Section VIII concludes the paper.

II. Unconstrained Planar Guidance Core

A. Relative Motion

Consider a stationary target at the origin and a pursuer moving at constant nominal speed $V > 0$ in a planar engagement. Let

$$r = \begin{bmatrix} r_x & r_y \end{bmatrix}^T, \qquad \gamma \in \mathbb{R} \tag{1}$$

denote the relative position and flight-path angle, respectively. The normalized lateral-acceleration command is

$$u = \frac{A}{A_{\max}}, \qquad |u| \leq 1. \tag{2}$$

The ideal acceleration-controlled guidance core is

$$\dot{r}_x = -V\cos\gamma, \qquad \dot{r}_y = -V\sin\gamma, \qquad \dot{\gamma} = \frac{A_{\max}}{V}u. \tag{3}$$

The closed-loop simulations pass the optimized command u_c through a first-order autopilot,

$$\dot{\gamma} = \frac{A_{\max}}{V}u_a, \qquad \tau_a\dot{u}_a = -u_a + u_c, \qquad |u_c| \leq 1, \tag{4}$$

where u_a is the actual normalized lateral acceleration. The ideal model (3) is recovered when $u_a = u_c = u$.

Assumption 1: The target is stationary, the pursuer speed is constant within each case, the command is piecewise constant and bounded, and the engagement is considered before the singular intercept instant. The heading angle is treated as an unwrapped real variable when differentiability is required.

III. Exact Bilinear Koopman Structure of the Guidance Core

A. Controlled Koopman Generator

For a controlled nonlinear system

$$\dot{x} = f_0(x) + uf_1(x), \tag{5}$$

the controlled Koopman generator acting on a smooth observable g is

$$\mathcal{L}_u g(x) = \nabla g(x)^T(f_0(x) + uf_1(x)) = \mathcal{L}_0 g(x) + u\mathcal{L}_1 g(x). \tag{6}$$

This expression is a generator representation for a controlled system. It should not be interpreted as a single autonomous Koopman operator independent of the input.

B. Exact Continuous-Time Bilinear Lift

Let the core state be

$$x_c = \begin{bmatrix} r_x & r_y & \gamma \end{bmatrix}^T. \tag{7}$$

For (3), define the analytic lifted state

$$z_c = \psi_c(x_c) = \begin{bmatrix} 1 & r_x & r_y & \gamma & \cos\gamma & \sin\gamma \end{bmatrix}^T. \tag{8}$$

The observable span generated by (8) is invariant under $\mathcal{L}_0$ and $\mathcal{L}_1$ because

$$\mathcal{L}_0 r_x = -V\cos\gamma, \qquad \mathcal{L}_0 r_y = -V\sin\gamma, \tag{9}$$

and

$$\begin{aligned} \mathcal{L}_1\gamma &= \frac{A_{\max}}{V}, \\ \mathcal{L}_1\cos\gamma &= -\frac{A_{\max}}{V}\sin\gamma, \\ \mathcal{L}_1\sin\gamma &= \frac{A_{\max}}{V}\cos\gamma. \end{aligned} \tag{10}$$

Therefore

$$\dot{z}_c = A_{0c}z_c + uB_{1c}z_c, \tag{11}$$

where the nonzero entries are

$$A_{0c}(2,5) = -V, \quad A_{0c}(3,6) = -V, \tag{12}$$

$$\begin{aligned} B_{1c}(4,1) &= \frac{A_{\max}}{V}, \\ B_{1c}(5,6) &= -\frac{A_{\max}}{V}, \\ B_{1c}(6,5) &= \frac{A_{\max}}{V}. \end{aligned} \tag{13}$$

Proposition 1: Under Assumption 1, the planar stationary-target guidance core (3) admits the exact finite-dimensional continuous-time bilinear Koopman representation (11) on the observable space spanned by (8).

Remark 1: The exact continuous-time bilinear representation is a structural result for the ideal core. Under zero-order-hold input, the exact discrete lifted map is $z_{c,k+1} = \exp(T_s(A_{0c} + u_k B_{1c}))z_{c,k}$, which depends nonlinearly on u_k. The online predictor below is therefore a task-oriented finite-dimensional approximation on the sampled operating domain.

IV. Angle-Only Terminal Guidance Formulation

A. Terminal Requirements Without Scheduled Impact Time

Let γ_f be the desired impact angle. The angle-only terminal requirements are

$$\min_t \|r(t)\| \leq r_c, \qquad |\gamma(t_{\mathrm{imp}}) - \gamma_f| \leq \varepsilon_\gamma, \tag{14}$$

where t_{imp} is the first capture time if the capture radius is reached, or the closest-approach time otherwise. The current simulations use either $r_c = 5$ m for the tuned 30° case or $r_c = 10$ m for nominal comparison cases, and $\varepsilon_\gamma = 3^\circ$. No equality constraint is imposed on t_{imp}.

B. Impact-Frame Error Coordinates

Define the desired impact-frame basis

$$e_f = \begin{bmatrix} \cos\gamma_f \\ \sin\gamma_f \end{bmatrix}, \qquad n_f = \begin{bmatrix} -\sin\gamma_f \\ \cos\gamma_f \end{bmatrix}. \tag{15}$$

The along-track distance, cross-track displacement, and heading error are

$$s = e_f^T r, \qquad y = n_f^T r, \qquad \theta = \gamma - \gamma_f. \tag{16}$$

The relative position can therefore be reconstructed from these two geometric error coordinates as

$$r = s e_f + y n_f, \qquad \|r\|^2 = s^2 + y^2. \tag{17}$$

For angle-only guidance, the along-track coordinate s is a distance-to-go variable in the desired impact frame, not a clock-time command. It is used as a progress coordinate and as part of the near-target terminal reference inside the receding-horizon problem.

The normalized prediction output is

$$x_e = \begin{bmatrix} \bar{s} & \bar{y} & \theta \end{bmatrix}^T, \qquad \bar{s} = \frac{s}{R_s}, \quad \bar{y} = \frac{y}{R_s}, \tag{18}$$

where R_s is the position scale. The nominal error dynamics are

$$\dot{s} = -V\cos\theta, \qquad \dot{y} = -V\sin\theta, \qquad \dot{\theta} = \frac{A_{\max}}{V}u. \tag{19}$$

With the first-order autopilot, u in (19) is replaced by u_a and the actuator state obeys (4).

V. Bilinear EDMD Predictor and Sequential Convex MPC

A. Lifted Dictionary and Identification

The analytic lift in Section III explains the bilinear structure of the ideal guidance core. The online controller uses a task-oriented lifted state built from the terminal-error coordinates and the autopilot state. In the large-angle implementation, the heading output is represented by $\sin\theta$ and $\cos\theta$ to avoid poor conditioning near large angle excursions. A representative lifted dictionary is

$$\begin{aligned} \psi(x_e, u_a) = [&1, \bar{s}, \bar{y}, \sin\theta, \cos\theta, u_a, \bar{s}\cos\theta, \bar{s}\sin\theta, \\ &\bar{y}\cos\theta, \bar{y}\sin\theta, u_a\cos\theta, u_a\sin\theta, \dots]^T, \end{aligned} \tag{20}$$

where $\bar{s}$ and $\bar{y}$ are the normalized components in (18). From sampled trajectories, the bilinear EDMD predictor is identified as

$$z_{k+1} = A_b z_k + B_0 u_k + u_k B_b z_k + w_k, \tag{21}$$

where w_k is the finite-dimensional prediction residual. With

$$\Omega_b = \begin{bmatrix} Z_- \\ U_- \\ U_- \odot Z_- \end{bmatrix}, \tag{22}$$

the ridge-regression estimate is

$$[A_b \; B_0 \; B_b] = Z_+ \Omega_b^T (\Omega_b \Omega_b^T + \rho I)^{-1}. \tag{23}$$

B. Sequential Convex Approximation

Direct MPC with (21) is nonconvex because of the product $u_k B_b z_k$. At each sampling instant, the controller uses a small number of sequential convex iterations. Given a nominal lifted trajectory $\bar{z}_{i|k}$, define

$$B_{\mathrm{eff},i} = B_0 + B_b \bar{z}_{i|k}. \tag{24}$$

The local prediction model is

$$z_{i+1|k} = A_b z_{i|k} + B_{\mathrm{eff},i} u_{i|k}. \tag{25}$$

The optimized sequence is damped between iterations to reduce numerical chattering caused by the bilinear approximation.

C. Quadratic Program

Let $\mathbf{U}_k = [u_{0|k}, \dots, u_{N-1|k}]^T$ and let the stacked predicted outputs be affine in the decision variables after linearization:

$$\mathbf{X}_k = c_k + M_k \begin{bmatrix} \mathbf{U}_k \\ \xi_k \end{bmatrix}. \tag{26}$$

The online optimization is

$$\begin{aligned} \min_{\mathbf{U}_k, \xi_k, s_f} \quad & \|\mathbf{X}_k - \mathbf{X}_k^{\mathrm{ref}}\|_Q^2 + \|\mathbf{U}_k\|_R^2 + \|D\mathbf{U}_k - d_{u,k}\|_{R_\Delta}^2 \\ & + \|z_k^m + \xi_k(z_k^p - z_k^m)\|_{Q_0}^2 \\ & + \lambda \xi_k^2 \|z_k^p - z_k^m\|^2 + \|s_f\|_{S_f}^2 \end{aligned} \tag{27}$$

subject to

$$-1 \le u_{i|k} \le 1, \qquad 0 \le \xi_k \le 1, \qquad s_f \ge 0, \tag{28}$$

the heading path constraint

$$|\gamma_f + \theta_{i|k}| \le \gamma_{\max}, \tag{29}$$

and optional acceleration-difference constraints

$$|u_{i|k} - u_{i-1|k}| \le \Delta u_{\max}. \tag{30}$$

The matrix D is the first-difference matrix, so the R_Δ term penalizes both the first move from the previous applied input and subsequent input changes.

The angle-only terminal constraint is imposed at the end of the prediction horizon rather than at a scheduled impact-time index:

$$\left\| \begin{bmatrix} s_{N|k}/R_s \\ y_{N|k}/R_s \\ \theta_{N|k} \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\| \le \varepsilon_N^{\mathrm{tight}} + s_f. \tag{31}$$

This terminal target encourages near-target capture with the desired terminal direction but does not require the target to be reached at a prescribed absolute time.

VI. Residual Tube Tightening

A. Multi-Step Residual Tube

Let $X_{h,j}^{\text{true}}$ and $X_{h,j}^{\text{pred}}$ denote the true and predicted terminal-error outputs at prediction step h for validation case j. The componentwise residual tube is estimated by

$$\rho_h = \alpha_\rho \, \text{quantile}_{0.99}\{|X_{h,j}^{\text{true}} - X_{h,j}^{\text{pred}}|\}_j, \tag{32}$$

where $\alpha_\rho = 1.2$ is an inflation factor. The tightened terminal tolerance at step h is

$$\varepsilon_h^{\text{tight}} = \varepsilon - \min\{\eta_{\max}\varepsilon, \rho_h\}, \tag{33}$$

with $\eta_{\max} = 0.8$. For the tuned 30° angle-only run, the terminal tube bound was

$$[9.731{\times}10^{-5},\ 9.991{\times}10^{-5},\ 1.242{\times}10^{-3},\ 1.182{\times}10^{-3}]^T, \tag{34}$$

for the $\sin\theta, \cos\theta$ output representation. The corresponding tightened terminal tolerance was

$$[4.027{\times}10^{-4},\ 4.001{\times}10^{-4},\ 5.109{\times}10^{-2},\ 2.741{\times}10^{-4}]^T. \tag{35}$$

B. Stability Interpretation

The residual tube is an empirical mechanism for connecting nominal predicted feasibility with true closed-loop performance. If the terminal lateral-error dynamics under a local feedback satisfy an input-to-state stability inequality with respect to the prediction disturbance, then tightening by (33) reduces the probability that finite-dimensional prediction error pushes the true state outside the terminal tolerance. The present implementation uses this tube as a numerical design tool; a computed invariant terminal set remains a topic for future work.

VII. Numerical Results

A. Simulation Setup

The main angle-only run starts from

$$r(0) = \begin{bmatrix}10000 & 0\end{bmatrix}^T \text{m}, \qquad \gamma(0) = -30^\circ, \tag{36}$$

with a desired impact angle of 30°. The nominal speed is 300 m/s, the stress speed is $0.97V$, $T_s = 0.1$ s, $A_{\max} = 100$ m/s^2, and $\tau_a = 0.10$ s. The main tuned setting uses $\Delta A_{\max} = 5$ m/s^2 for applied acceleration change, $R_\Delta = 1.5$, and a maximum simulation time of 50 s. The prediction model is trained from sampled stationary-target trajectories and the online controller is solved as a sequence of convex quadratic programs.

B. Prediction Accuracy

The exact bilinear lift of the ideal acceleration-controlled core was checked by integrating the lifted and original systems with the same numerical scheme. In the tuned 30° run, the maximum consistency error was 2.39×10^{-11}, confirming the exact continuous-time closure of the core model. The learned autopilot-augmented predictor was then evaluated on validation data.

TABLE I
Prediction Error on the Tuned Angle-Only Test Set

State	1-step	1-step roll	Linear multi-step	Bilinear roll
s/R_s	$5.77{\times}10^{-6}$	$3.95{\times}10^{-2}$	$3.42{\times}10^{-2}$	$3.85{\times}10^{-5}$
y/R_s	$4.07{\times}10^{-6}$	$2.81{\times}10^{-2}$	$2.45{\times}10^{-2}$	$2.72{\times}10^{-5}$
$\sin\theta$	$3.04{\times}10^{-3}$	$3.71{\times}10^{-1}$	$3.20{\times}10^{-1}$	$3.33{\times}10^{-4}$
$\cos\theta$	$2.97{\times}10^{-3}$	$3.55{\times}10^{-1}$	$3.03{\times}10^{-1}$	$3.19{\times}10^{-4}$

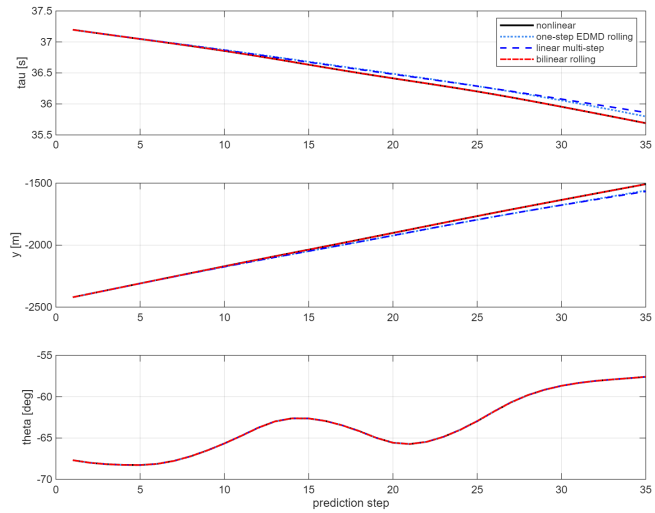


Fig. 1. Prediction comparison for the tuned angle-only run. The bilinear rolling predictor gives substantially lower long-horizon normalized error than rolling one-step EDMD or direct linear multi-step EDMD on the same validation set.

TABLE II
Main 30° Angle-Only Case With 5 m Capture Radius

Case	Captured	Impact time	Miss	Angle error
Nominal	Yes	37.0 s	3.048 m	0.109°
Stress	Yes	38.0 s	4.150 m	0.121°

C. Angle-Only Capture Performance

Table II reports the main 30° angle-only case with a 5 m capture radius. The nominal case entered the capture set at 37.0 s with a 3.048 m miss distance and a 0.109° impact-angle error. The speed-perturbed stress case entered the capture set at 38.0 s with a 4.150 m miss distance and a 0.121° angle error. Both cases had no QP failures.

Fig. 2 shows that the trajectory reaches the target neighborhood while aligning the heading with the desired impact direction. The command and first-order-autopilot acceleration remain smooth because the input-difference penalty and acceleration-change bound suppress high-frequency command variation.

D. Nominal 30° and 45° Cases

The earlier nominal angle-only tests used a 10 m capture radius and did not prescribe terminal time. Table III reports that both the 30° and 45° commands satisfied the capture and impact-angle requirements. The actual impact times were 22.1 s and 22.9 s, respectively.

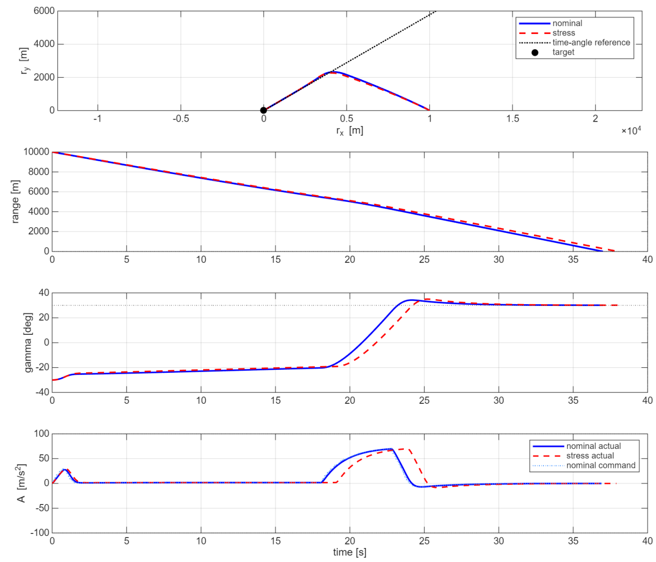


Fig. 2. Closed-loop trajectory, range history, heading response, and lateral acceleration for the tuned 30° angle-only case. The actual impact time is an output of the closed-loop trajectory rather than a prescribed terminal constraint.

TABLE III
NOMINAL IMPACT-ANGLE-ONLY CASES WITH 10 M CAPTURE RADIUS

γ_f	Impact time	Miss	Angle error
30°	22.1 s	6.945 m	−0.107°
45°	22.9 s	7.280 m	−0.404°

E. Acceleration-Change Constraint Scan

The acceleration-change constraint was scanned to examine the role of command smoothing. Table IV shows that an intermediate value, $\Delta A_{\max} = 5$ m/s^2, gave the best nominal and stress capture performance in this set. Removing the constraint caused larger acceleration variation and did not satisfy the 5 m nominal capture requirement. Over-tightening the constraint also degraded capture performance by limiting terminal maneuverability.

F. Large-Angle Boundary Cases

Large impact-angle commands expose the current reachable-envelope limitation. A 60° balanced 5 s-horizon case with $\Delta A_{\max} = 30$ m/s^2 reached 8.187 m nominal miss and $-1.380°$ angle error, but it did not satisfy the stricter 5 m capture requirement and produced many QP failures. A 10 s move-blocking variant reduced trajectory instability relative to unconstrained long-horizon optimization, but still missed the 5 m capture set and produced a larger angle error. These cases indicate that the present angle-only Koopman-MPC design is reliable for moderate impact angles but not yet a complete large-angle guidance solution.

VIII. DISCUSSION

The results support three bounded conclusions. First, the exact bilinear Koopman derivation for the ideal Cartesian guidance core provides a structural reason to prefer a bilinear lifted predictor over a purely linear input predictor. The validation data support this point: the bilinear rolling predictor substantially reduced long-horizon normalized prediction error in the tuned angle-only case. Second, removing the prescribed impact-time constraint makes the terminal task more reachable for the tested nominal cases. The main 30° case satisfied a 5 m capture radius under both nominal and speed-perturbed conditions, and the 30° and 45° nominal cases satisfied a 10 m capture radius without scheduling a terminal clock time. Third, command smoothing is not merely a cosmetic addition. In the acceleration-change scan, moderate smoothing improved the miss distance, whereas both insufficient and excessive smoothing degraded capture.

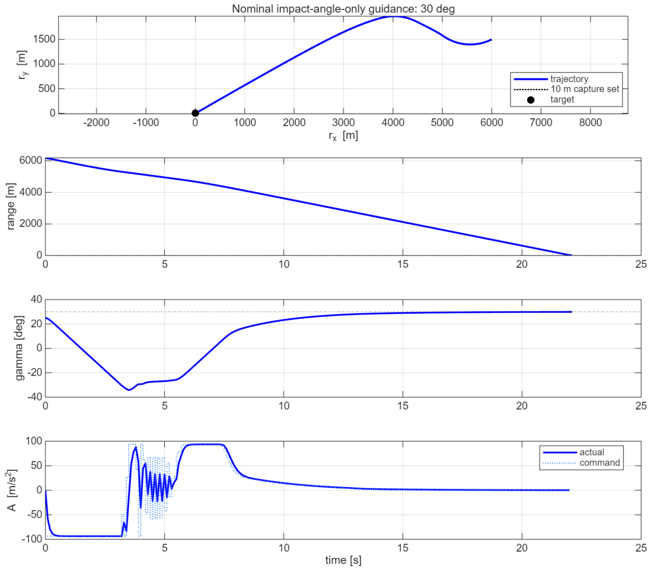
The same results also show the limits of the current design. The large-angle cases did not satisfy the strict 5 m capture radius and produced many QP failures, especially when the angle target and prediction horizon demanded strong terminal curvature. This behavior suggests that large-angle interception needs additional reachability-aware reference shaping, better terminal-set design, or a more reliable large-domain predictor. Thus, the present manuscript should not claim a general solution for arbitrary large impact-angle commands.

Compared with the fixed-time formulation, the angle-only formulation gives a cleaner current paper story. Fixed-time experiments revealed that feasible terminal-time windows can be narrow and highly sensitive to tuning. Treating impact time as an outcome avoids over-constraining the terminal problem and lets the paper focus on the part currently supported by evidence: Koopman-MPC prediction and QP-based receding-horizon guidance for impact-angle-constrained capture.

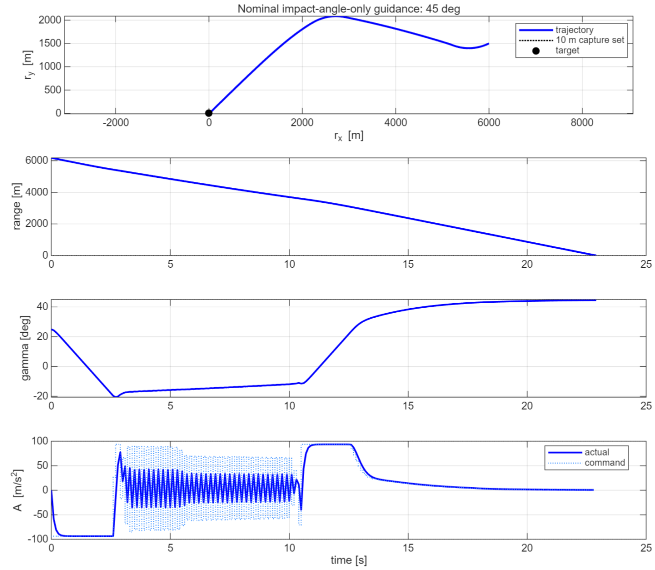
The study remains nominal. The plant model uses stationary-target geometry, constant speed within each case, and a first-order autopilot. It does not yet include target maneuvers, aerodynamic drag, higher-order actuator dynamics, sensor field-of-view constraints, or formal recursive-feasibility guarantees. Future work should compute a terminal invariant set for the angle-only coordinate, enlarge the large-angle training domain, and introduce reachability-aware terminal reference scheduling before returning to fixed impact-time constraints.

IX. CONCLUSION

This paper presented a tube-tightened bilinear Koopman-MPC framework for nominal planar impact-angle guidance without prescribed impact time. The method combines an exact continuous-time bilinear Koopman structure for the ideal stationary-target guidance core with a task-oriented bilinear EDMD predictor, first-order-autopilot augmentation, sequential convex quadratic programming, terminal slack variables, input-difference regularization, and validation-data residual-tube tightening. In the main 30° angle-only case, the controller achieved a 3.048 m nominal miss distance and a 0.109° impact-angle error within a 5 m capture radius; the speed-perturbed stress case also satisfied the capture condition. Nominal 30° and 45° cases satisfied a 10 m capture-radius requirement. Acceleration-rate scans and large-angle tests show that the method remains tuning-sensitive and incomplete for aggressive impact-angle commands. The appropriate



(a) 30° impact-angle command.



(b) 45° impact-angle command.

Fig. 3. Nominal angle-only flight simulations for 30° and 45° impact-angle commands. Both cases satisfy the 10 m capture-radius condition and the 3° impact-angle tolerance without prescribing impact time.

TABLE IV
ACCELERATION-CHANGE CONSTRAINT SCAN FOR THE 30° ANGLE-ONLY CASE

$\Delta A_{\max}$ [m/s^2]	Nominal captured	Nominal miss [m]	Nominal angle [deg]	Stress captured	Stress miss [m]	QP failures
∞	No	13.489	-0.081	Yes	3.213	0/0
15	No	10.755	-0.086	No	10.707	0/0
10	No	8.208	-0.070	No	8.095	0/0
5	Yes	3.048	0.109	Yes	4.150	0/0
3	No	17.709	0.959	No	19.816	0/0

TABLE V
REPRESENTATIVE LARGE-ANGLE BOUNDARY CASES

Case	Horizon	Miss [m]	Angle error	Failures
60° balanced	5 s	8.187	$-1.380°$	163
60° move blocking	10 s	9.076	$-9.449°$	146
80° quick test	5 s	11.861	$-1.285°$	290

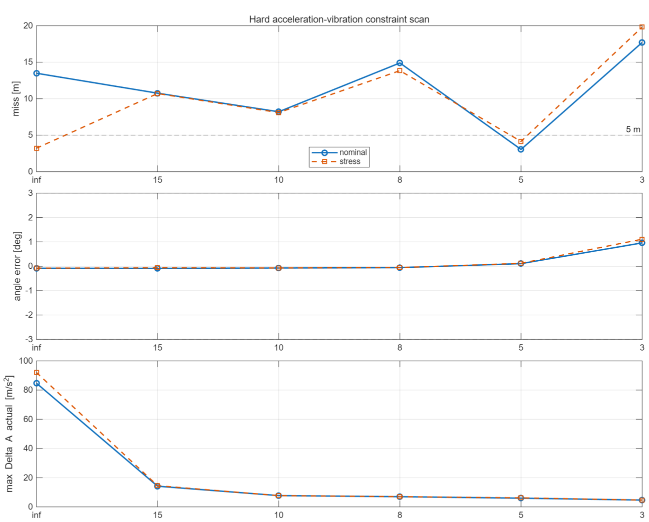


Fig. 4. Acceleration-change constraint scan. Moderate smoothing improves capture in this tuning set, while either removing the bound or making it too tight degrades the terminal miss distance.

conclusion is therefore bounded: removing the fixed impact-time constraint yields a more credible and better-supported Koopman-MPC guidance formulation for the present data and simulations.

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